IS1204
Fundamentals of Software Engineering

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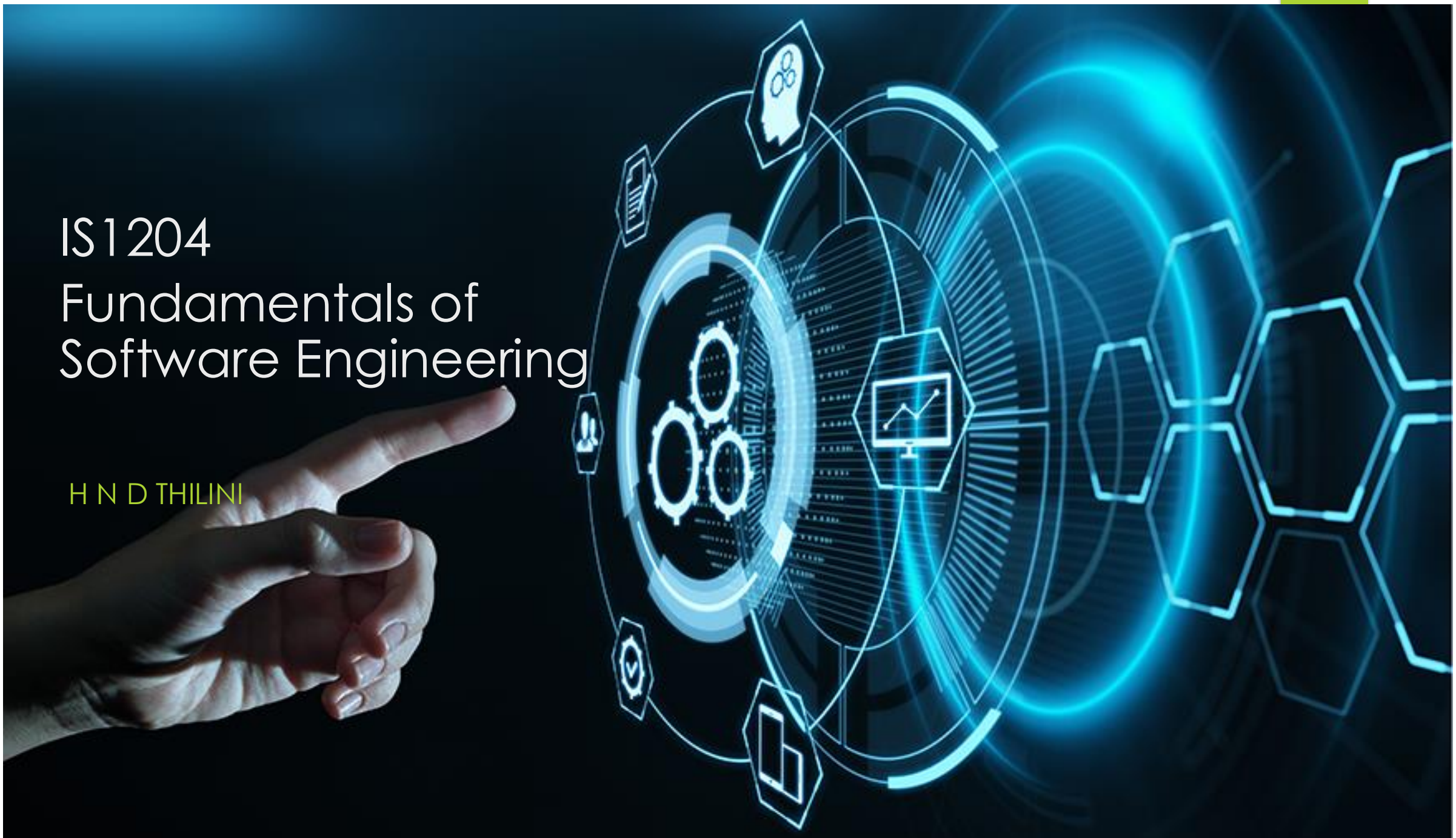
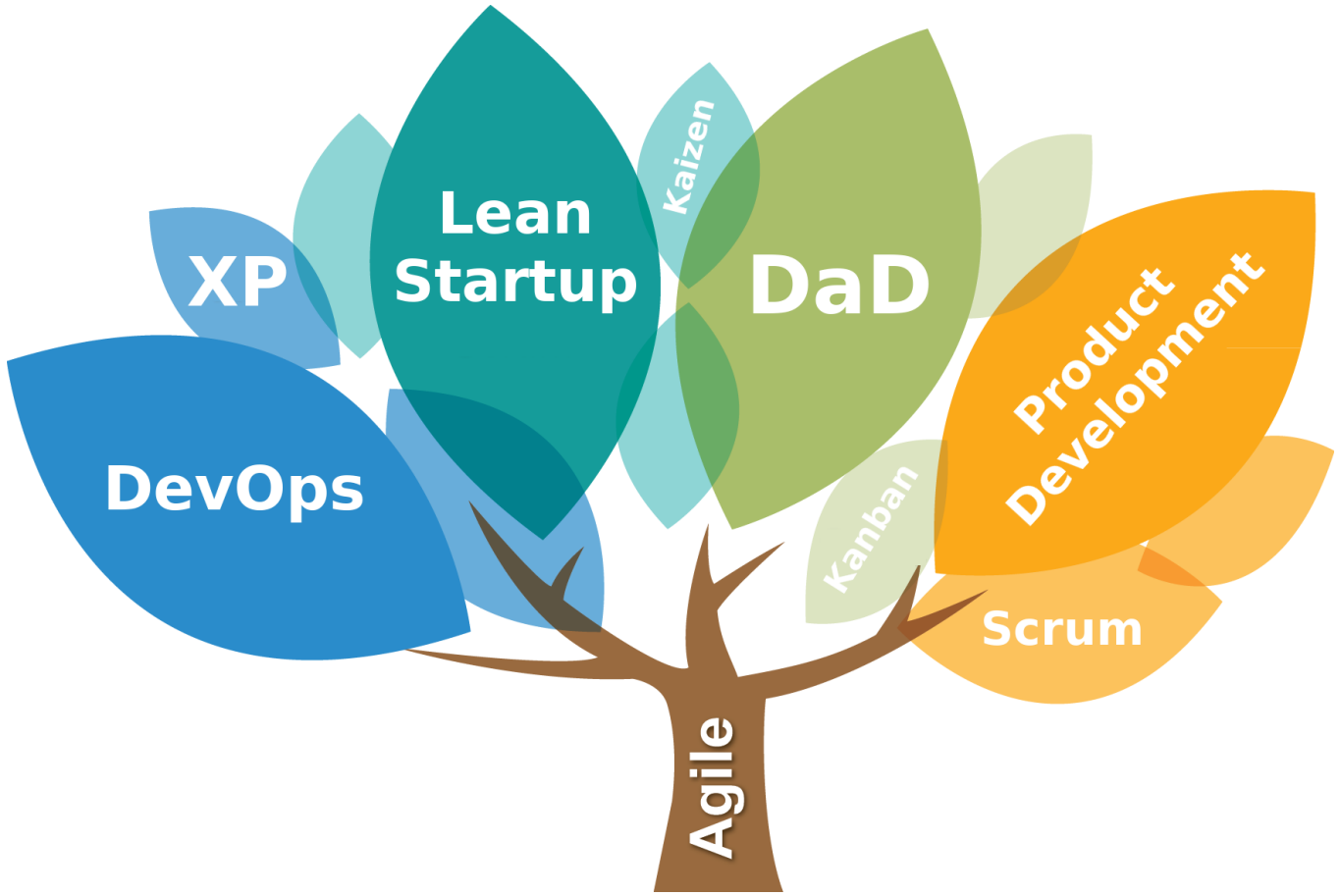




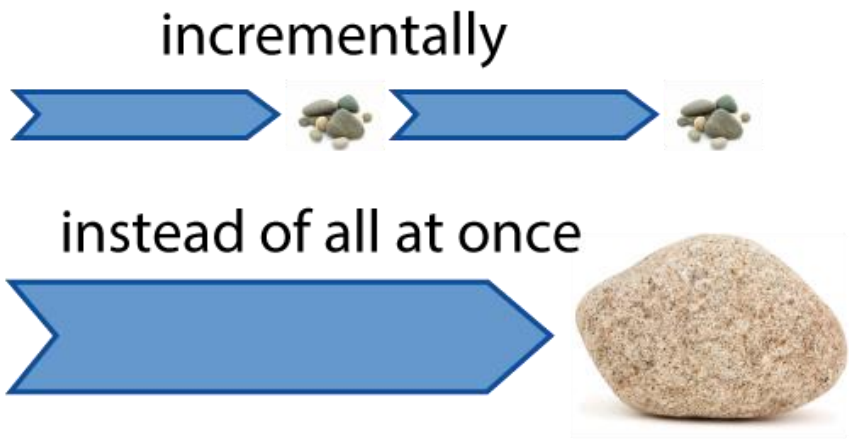
Agile Software Development

Learning Outcomes

- Define agile software development
- Understand the rationale for agile software development methods and agile manifesto.
- Compare and contrast agile and plan-driven development.
- Identify the key practices in extreme programming and how these relate to general practices of agile methods.
- Understand Scrum, Kanban, Lean approach to agile software development.

An Introduction to Agile?

- Why does it exist?
- What problems does it solve?
- What is it?

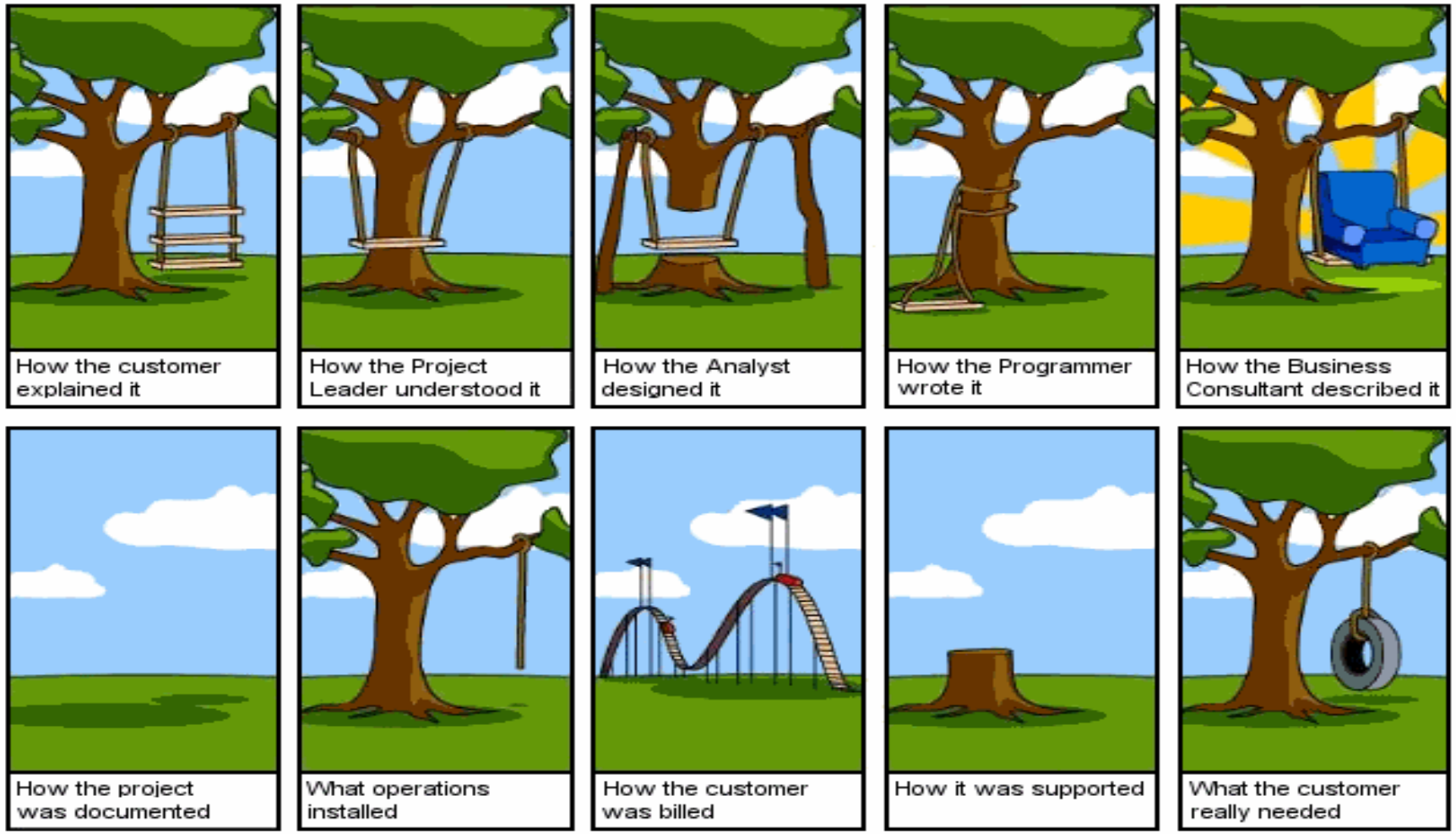


- Agile is a **time boxed**, **iterative approach** to software delivery that builds software **incrementally** from the start of the project, instead of trying to deliver it all at once near the end.

Most software developments fail

- Software developments projects being **cancelled** every now and then
- Software projects are being considered as **failures** by those who initiated it.
- One in every two projects **exceed its budget** by 200%.

WHY / What is wrong?



WHY? Cont….

- Do not meet the need of the user.
- Deadline rush.
- Less number of features delivered.
- Poor interfaces.

Success is not just functions anymore!!!

LOT more EXPECTATIONS - Users

- Speed Delivery
- Accuracy
- Features vs Usability

What causes most of the software developments fail?????????

Traditional Waterfall Method

Waterfall model build-upon Assumptions

- Requirements can be entirely predicted upfront.
- Each phase of the lifecycle can be perfected before moving to the next.
- Timeframes and budgets can be predicted up front.
- The feedback from the real user is not so valuable.

UNREALISTIC

Causes for Failures

- Risk is bundled!

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Causes for Failures

- Poor communication!
- Feedback is obtained at the end of the project

Beginning → End

Can you meet the client requirement at this stage????

Agile methods

Topics to Discuss

1. What is Agile

   Introduction, Why Agile came into play, History of Agile

2. Agile Manifesto

   4 core values, 12 principles of agile

3. Agile Development Life Cycle

   How it is done, Different stages, Central activities

4. Advantages & Disadvantages of Agile

5. Agile Vs Traditional software development approaches

   Comparison with other approaches, What types of projects is agile suitable for

Agile Practices

1. Incremental planning
2. Collective ownership
3. Continuous integration
4. On-site customer

Group 2

5. Using User Stories **Group 3**

6. Pair programming **Group 4**

7. Refactoring
8. Simple design
9. Small releases
10. Sustainable pace

Group 5

11. Test-Driven development
12. Behaviour-Driven development

Group 6

Rapid software development

- Rapid development and delivery is now often the most important requirement for software systems
  - Businesses operate in a fast–changing requirement, and it is practically impossible to produce a set of stable software requirements
  - Software must evolve quickly to reflect changing business needs.
- Plan-driven development is essential for some types of system but does not meet these business needs.
- Agile development methods emerged in the late 1990s whose aim was to radically reduce the delivery time for working software systems

Agile methods

- Dissatisfaction with the overheads involved in software design methods of the 1980s and 1990s led to the creation of agile methods.
- These methods:
  - Focus on the **code** rather than the design
  - Are based on an **iterative approach** to software development
  - Are intended to **deliver working software quickly** and evolve this quickly to meet changing requirements.
- The aim of agile methods is to reduce overheads in the software process (e.g. by limiting documentation) and to be able to respond quickly to changing requirements without excessive rework.

AGILE MANIFESTO

CUSTOMER COLLABORATION over contract negotiation

INDIVIDUALS AND INTERACTIONS over processes and tools

RESPONDING TO CHANGE over following a plan

WORKING SOFTWARE over full documentation

www.softwaretestingclass.com

Agile manifesto cont….

- *We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:*

Individuals and interactions	*over*	*processes and tools*
Working software	*over*	*comprehensive documentation*
Customer collaboration	*over*	*contract negotiation*
Responding to change	*over*	*following a plan*

- *That is, while there is value in the items on the right,* we value the *items on the left more*.

12 Agile Principles

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
4. Business people and developers must work together daily throughout the project.
5. Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
7. Working software is the primary measure of progress.
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. Continuous attention to technical excellence and good design enhances agility.
10. Simplicity--the art of maximizing the amount of work not done--is essential.
11. The best architectures, requirements, and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

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1.Early and continuous delivery of valuable software

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2. Welcome Changing Requirements

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3. Deliver working software frequently

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4. Business people and developers must work together

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5. Build projects around motivated individuals

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6. Face-to-face conversation

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8. Promote sustainable development

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9. Technical excellence and good design.

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10. Maximizing the amount of work not done.

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What is Agile Software Development?

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Is Agile better?

Feedback Point

- Corrections
- Modifications
- Re-prioritizing
- Usability analysis

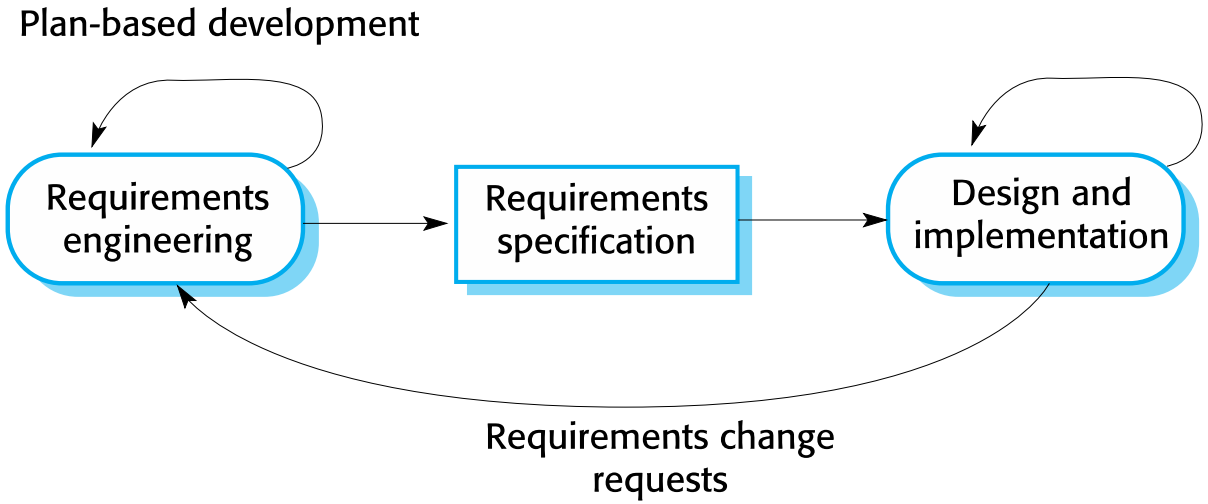
Agility

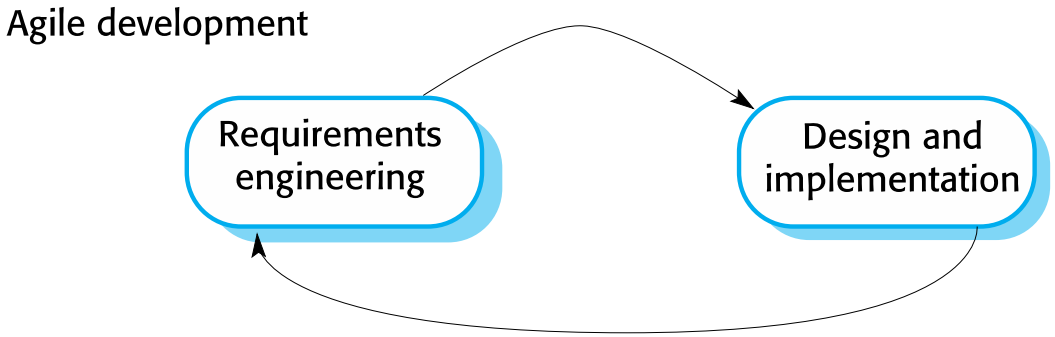
- Decrease Risk
- More Feedback
- Less confusion
- More Satisfaction

What is Agile software development cont….

- Program specification, design and implementation are inter-leaved
- The system is developed as a series of versions or increments with stakeholders involved in version specification and evaluation
- Frequent delivery of new versions for evaluation
- Extensive tool support (e.g. automated testing tools) used to support development.
- Minimal documentation – focus on working code

Plan-driven and agile development





Plan-driven and Agile development

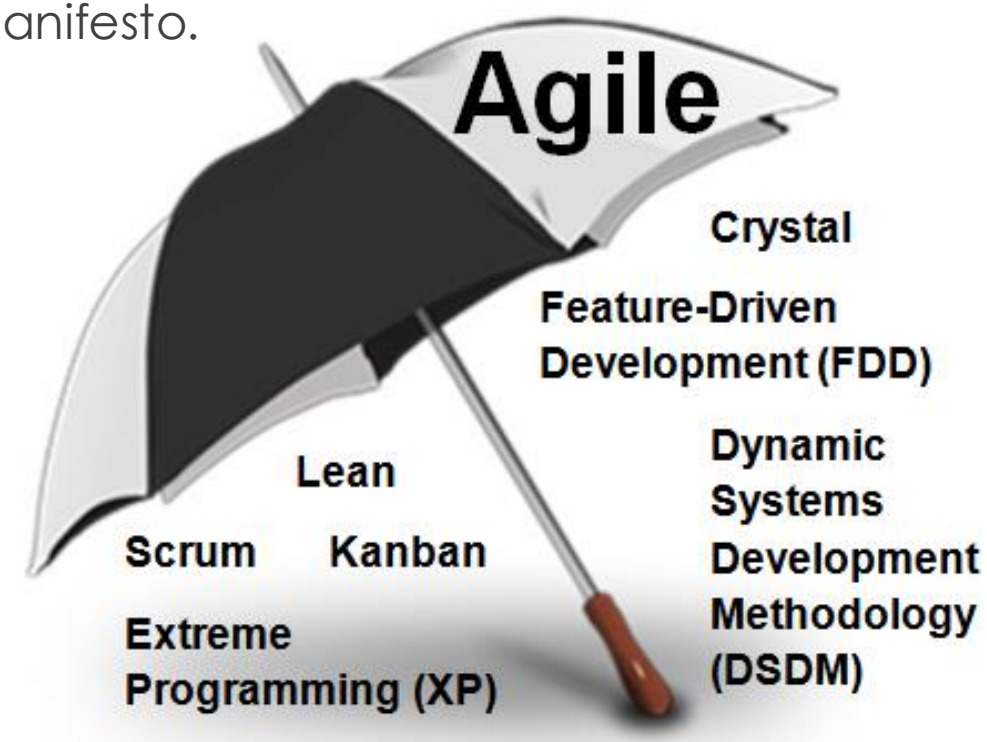
- Plan-driven development
  - A plan-driven approach to software engineering is based around separate development stages with the outputs to be produced at each of these stages planned in advance.
  - Not necessarily waterfall model – plan-driven, incremental development is possible.
  - Iteration occurs within activities.
- Agile development
  - Specification, design, implementation and testing are inter-leaved and the outputs from the development process are decided through a process of negotiation during the software development process.

What is Agile Software Development?

- A mindset

What is Agile Software Development?

- An umbrella term for a set of methods and practices based on the values and principles expressed in the Agile Manifesto.



Agile Software Development - Successes

- Small or medium sized product development.
- Custom system development within organization
- Experiments in using agile approaches for critical systems engineering.
  - Security, safety, and dependability analysis.
  - Need significant modifications before applying.

Drawbacks / Difficulties

- Custom representatives are subject to other pressures and cannot take full part in the software development.
- Individual team members may not have suitable personalities for the intense involvement / teamwork.
- Prioritizing changes can be extremely difficult when there are lot of stakeholders.
- Under pressure from delivery schedules, the team may not have time to carry system simplifications.
- Cultural change – team members may find it hard to change the practice (informal / defined by team itself)

Plan driven vs Agile

- Detailed design and specification before moving to implementation? Plan-driven approach
- Incremental delivery a realistic? Agile
- The system being development is large? Plan-driven approach
- Real-time system with complex timing requirements? Plan-driven approach
- Expected system lifetime?
- Technology?
- Development team organization?

Plan driven vs Agile

- Cultural issues that may affect the system development?
- Skills of designers and developers?
- Subjected to any external regulations?

An Executable Software that meets the need and does useful things for the individual user or the organization.

Exercise 1

- Explain why the rapid delivery and deployment of new systems is often more important to businesses than the detailed functionality of the systems.

Exercise 2

- When would you recommend the use of agile for developing a software system?

Exercise 3 (Homework)

- Explain how the principles underlying agile methods lead to the accelerated development and deployment of software.